Any two ordinals are comparable by inclusion: alpha subset of beta or beta subset of alpha

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the three-case inclusion argument and the closing trichotomy citation are verbatim in the span.

Editorial note. The label ordinalsaresubsets names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Trichotomy for ordinal membership. Same unit, \ollabel{ordtrichotomy} (source line 477). Final line of the proof (596): the three membership cases exhaust all possibilities for alpha, beta.
- Transitivity of ordinals. The unit's standing ordinal facts invoked in-line at 592-594 ('as beta is transitive', 'Similarly'). Turns membership of an ordinal into inclusion of its elements.

Source reference. Original reference ordtrichotomy: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 477–477. Prerequisite pointer invoked at proof line(s) 596; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *$\alpha \subseteq \beta \vee \beta \subseteq \alpha$, for any ordinals α, β.*

Proof. If $\alpha \in \beta$, then $\alpha \subseteq \beta$ as β is transitive. Similarly, if $\beta \in \alpha$, then $\beta \subseteq \alpha$. And if $\alpha = \beta$, then $\alpha \subseteq \beta$ and $\beta \subseteq \alpha$. So by ordtrichotomy we are done. □